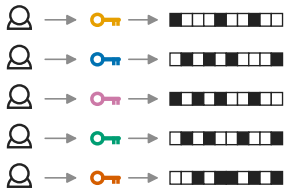


Fresh keys

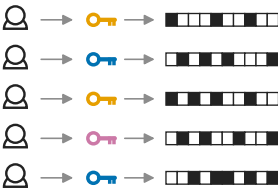
one key per source record



no transform reuse

Recurring key pool

k transforms; slot labels hidden

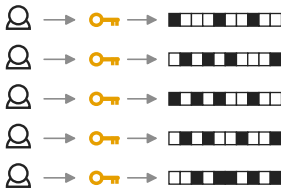


same colour = same transform

reuse across train and test

Single shared key

k = 1; key value still hidden



shared-transform baseline

Chance-compatible results

ideal invariance needs assumptions

Multi-record gains observed

boundary depends on dataset and split

Strong linkage observed

hidden does not mean unlearnable

Protocol

Disjoint train / validation / test identities; one held-out gallery image per identity
8 nested sets per identity; 3 model seeds; identity-clustered bootstrap intervals